

CodeAlpha Internship Project Report

Project 3: Object Detection and Tracking

Objective: Build a real-time object detection and tracking system using OpenCV and a pre-trained YOLO model.

Tools: Python, OpenCV, YOLOv8/YOLOv5, NumPy.

Features:

- Webcam/video input
- Detect objects
- Draw bounding boxes
- Assign tracking IDs
- Real-time display

Algorithm:

1. Capture frames.
2. Run YOLO detection.
3. Track objects across frames.
4. Display labels and IDs.

Sample Code:

```
from ultralytics import YOLO
model=YOLO('yolov8n.pt')
results=model.predict(source=0)
```

Expected Output: Live video with detected objects enclosed in bounding boxes and tracking IDs.

Future Scope: Face recognition, traffic monitoring, surveillance, smart retail analytics.

Conclusion: This project demonstrates computer vision techniques for real-time object detection and tracking.